Dynamic Games with Incomplete Information

Sharp history-based identification and certified inference

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Abstract

For arbitrary finite known perfect-recall trees, we state restrictions under which incomplete-information games admit informative, sharp, and computationally certified inference. A calibrated measurement channel on complete terminal histories recovers the latent law. Positive history mass then identifies behavior and beliefs and reduces the entire compatible parameter set to an explicit affine slice of a joint parameter polytope. A rational projection and linear program give finite-sample whole-set confidence, conditioned mean-square parameter bounds, expected diameter bounds, and an actual feasible Bayes assessment with a certificate for every continuation deviation. General policy ranges retain all sequential equilibria at a shared structural parameter and have exact real-algebraic endpoint certificates. For two-player zero-sum utility values, sequence-form linear programs and a coupling bound give polynomial explicit-tree computation and stable confidence intervals. A tie example proves that the same contraction claim cannot hold for arbitrary policy outcomes, at any fixed confidence level. Private histories are retained throughout. The algorithms are proved; a full QE or LP execution is not claimed.

1 Declared class, observation design, and scope

The canonical entry asks which restrictions make identified and counterfactual sets informative, and requests verified computation, approximation, sampling confidence, and an explicit policy-selection convention. The results below concern the following declared class. They do not assert identification under every unrepresented primitive family.

The input is an arbitrary finite known extensive-form tree with perfect recall. It may be the complete history tree of a finite-horizon game with persistent private states, correlated signals, and arbitrary observation histories. Strategic information sets and their common nonempty action sets are known. No public Markov reduction is made. Let Z be the terminal-history set, $L = |Z|$, N_{tr} the number of nodes, and H the maximum number of edges on a terminal path. Each z records the *complete* latent history, including private signals and Nature events, not merely public outcomes.

The structural parameter is $\gamma = (\alpha, \beta) \in \Gamma \subset \mathbb{R}^d$, where Γ is a nonempty compact rational polytope and d is the ambient coordinate count. Joint restrictions on α, β are retained. Empty parameter domains are detected and reported separately. The vector of all included Nature-edge probabilities is

$$c(\alpha) = c^0 + B\alpha, \qquad u_{i,z}(\beta) = u^0_{i,z} + \varphi_{i,z} \cdot \beta. \tag{1.1}$$

All displayed coefficients are known and rational. On Γ, Nature rows are proper probability vectors and every included edge is uniformly positive. Structurally absent edges are omitted;

strategic off-path actions are retained. Repeated transition and signal kernels are imposed by repeated rows or primitive equalities, not fitted as unrelated probabilities.

The known discount factor of the underlying dynamic game is denoted δ_0, to distinguish it from the unknown payoff vector β. Discounted stage-utility features are summed along a terminal history to obtain (1.1). Cardinal units, known utility entries, and any normalization are fixed as primitives. Additive or scale directions not removed by those restrictions remain unidentified. Let d_β be the dimension of β, and suppose

$$\|\beta\|_1 \le B_1, \quad U_0 = \max_{i,z} |u^0_{i,z}|, \quad F = \max_{i,z,j} |\varphi_{i,z,j}|, \quad U = U_0 + FB_1. \tag{1.2}$$

Empty maxima are zero. Thus $|u_{i,z}(\beta)| \le U$ on Γ.

After history z, the researcher observes one of K cells through a known column-stochastic channel $M(o \mid z)$. This calibrated post-play measurement is conditionally determined by z; it changes neither players' information nor their payoffs during play. Calibration aligns latent labels with their transition and payoff features. If q is the terminal law, the observed law is

$$p = Mq. \tag{1.3}$$

The generic exact method allows rank-deficient M and zero components of q. The informative branch assumes M has full column rank and the recovered population q is strictly positive. Quantitative statements additionally use a declared known rational $\lambda > 0$ with $q_z \ge \lambda$, necessarily $L\lambda \le 1$.

Independent markets share one baseline assessment, with independent Nature and behavioral randomization across markets. An observed selection stratum can be analyzed separately. An unobserved mixture over different assessments is a different sampling model. The sample size is n throughout this paper.

For each declared policy ℓ, another known finite perfect-recall tree satisfies the same compact, affine, positive-Nature assumptions at the *same* γ. A declared scalar outcome has the form

$$W_\ell(\gamma, \sigma) = \sum_{z \in Z_\ell} q^\ell_z(\gamma, \sigma) w_{\ell,z}(\gamma), \tag{1.4}$$

where $w_{\ell,z}$ are known rational polynomials. This includes probabilities, counts, and calibrated affine utility or welfare statistics. The attainment assertions concern these continuous outcomes. Vector targets use the joint image, not a Cartesian product of coordinate ranges.

Exact algorithms require rational or effectively represented algebraic numerical inputs, including the channel and numerical population law. Population statements also make sense for fixed known real inputs; an arbitrary unencoded real is not finite computational input. The polynomial-bit claims use rational encodings. The complete tree and channel may be exponentially large in a succinct horizon description.

The prior catalogue attempt [4] already proves the fixed-positive-Nature exact equilibrium reduction, compactness, sharp policy projections, endpoint bisection, and whole-set sampling coverage. We reproduce those arguments and attribute them. The additional content is the general informative inverse and affine fiber, its quantitative statistical and actual-assessment certificates, and qualified stable policy-value inference. Classical sequential equilibrium [1], real quantifier elimination [2], sequence form [3], and rational linear programming are named inputs.

2 Exact sequential equilibrium, existence, and compactness

Fix a baseline or policy tree. Let $\mathcal{B}$ be its product of behavioral simplexes, $\mathcal{B}^+$ its strictly positive part, and $\mathcal{U}$ the product of belief simplexes over the nodes of each information set. Write

$$R_h(\gamma, b) = \prod_{\text{edges on the prefix of } h} \text{their Nature or behavioral probabilities},$$

$$R_I(\gamma, b) = \sum_{h \in I} R_h(\gamma, b).$$

For $b \in \mathcal{B}^+$, every R_I is positive. Its Bayes graph is

$$b \in \mathcal{B}^+, \quad \nu \in \mathcal{U}, \quad R_I(\gamma, b) > 0, \quad \nu_{I,h} R_I(\gamma, b) = R_h(\gamma, b) \quad (h \in I).$$

The following formula defines consistency:

$$\forall r > 0 \ \exists (b, \nu) \text{ in this Bayes graph} : \|b - \sigma\|_2^2 + \|\nu - \mu\|_2^2 < r^2. \tag{2.1}$$

Here γ and all game primitives are held fixed. One common pair (b, ν) serves every information set. The witnesses need not be equilibria or rational.

For player i, information set I, and complete pure continuation plan a_i, let $V_{i,I}(\gamma, a_i, \sigma_{-i}; \mu)$ be the terminal-utility sum over suffix paths starting at $h \in I$, weighted by $\mu_{I,h}$. The corresponding prescribed-continuation value is $V_{i,I}(\gamma, \sigma; \mu)$. These are finite polynomials. Require the legitimate-assessment simplex conditions and

$$V_{i,I}(\gamma, \sigma; \mu) \geq V_{i,I}(\gamma, a_i, \sigma_{-i}; \mu) \quad \text{for every } i, I, a_i. \tag{2.2}$$

Their conjunction with (2.1) is denoted $\mathrm{SE}(\gamma, \sigma, \mu)$.

Lemma 1 (Exact fixed-game predicate)**.** *The predicate* SE *describes exactly the sequential-equilibrium assessments, including consistent off-path beliefs.*

Proof. A converging consistency sequence supplies a witness for every $r > 0$. Conversely, choose witnesses for $r = 1/m$. They give a single sequence of completely mixed profiles with jointly convergent Bayes beliefs in the same game.

Perfect recall prevents a path from revisiting an information set. Given any behavioral continuation deviation, independently draw in advance its action at each own information set. Every terminal suffix then has the same product probability as under behavioral play. The deviation's value is consequently a convex combination of values of the pure continuation plans in (2.2), and is no larger than the prescribed value. Necessity is immediate because pure plans are admissible. This proves full sequential rationality and the claim. □

All conditions are first-order ordered-field formulas. The radius in (2.1) is a real variable, not a quantifier over sequences. Pure continuation plans are finite and can be enumerated; alternatively behavioral deviations can be universally quantified using the same polynomial suffix sums. Real quantifier elimination therefore makes the exact relation effectively semialgebraic. This observation alone is not an unrestricted efficiency claim.

Lemma 2 (Nonempty equilibrium fibers)**.** *Every fixed admitted game has a sequential equilibrium.*

Proof. When there are no strategic information sets, the empty assessment suffices. Otherwise choose $0 < \varepsilon < 1/\max_I |A(I)|$ and require each action probability to be at least ε. Give each

information set one decision agent with the original player's unconditional utility. Parameterize its behavior by

$$\sigma_{I,a} = \varepsilon + (1 - |A(I)|\varepsilon)x_{I,a}, \qquad x_I \text{ in the ordinary simplex.}$$

An agent's payoff $H_A(x)$ is continuous and affine in its own block. To prove an agent equilibrium exists, put

$$g_{A,a} = \max\{0, H_A(a, x_{-A}) - H_A(x)\}, \qquad x_{A,a} \longmapsto \frac{x_{A,a} + g_{A,a}}{1 + \sum_b g_{A,b}}.$$

This continuous map of the compact convex product of simplexes has a fixed point by Brouwer's theorem. If $G_A = \sum_b g_{A,b} > 0$, the fixed-point equation gives $g_{A,a} = x_{A,a} G_A$. Every supported pure action would strictly improve $H_A(x)$, contradicting its expression as their weighted average. Thus $G_A = 0$ for every agent.

Let σ^ε be the resulting constrained agent equilibrium and μ^ε its Bayes beliefs. Every information set has positive reach. A unilateral agent change alters unconditional utility by that reach probability times its conditional local payoff change. The prescribed choice is therefore conditionally optimal among ε-constrained distributions.

Perfect recall gives all nodes of an own information set J the same remembered own sequence of earlier information sets and actions. The product of own earlier action probabilities is common to those nodes and cancels from Bayes' rule. With opponents and Nature fixed, the conditional belief at J is independent of one's earlier positive behavior. Moreover, if J follows I on one path, all its nodes remember I and lie in its continuation.

Start with any constrained full continuation from I. Replace its own decisions by σ^ε in decreasing order of remembered own-decision length. At a replacement at J, later own decisions already agree with σ^ε, and the conditional belief is μ^ε_J by the cancellation just proved. Local optimality makes the payoff change nonnegative; its contribution at I is multiplied by the probability of reaching J. This proves optimality against every constrained continuation.

Let $\varepsilon_m \downarrow 0$ and take a convergent subsequence of the compact assessment product. Its limit is consistent by construction. For an arbitrary unconstrained continuation δ_i, use at each own information set the constrained distribution $\varepsilon_m + (1 - |A(J)|\varepsilon_m)\delta_{i,J,a}$. It converges to δ_i. Passing to the limit in the finite polynomial continuation inequalities proves (2.2). □

The specially chosen perturbed equilibria in this existence proof are not imposed on the witnesses of (2.1).

Lemma 3 (Compact joint graph)**.** *The graph*

$$\mathcal{S} = \{(\gamma, e) : \gamma \in \Gamma,\ e = (\sigma, \mu),\ \mathrm{SE}(\gamma, e)\}$$

is compact and semialgebraic, including at strategically off-path information sets.

Proof. Semialgebraicity follows from Lemma 1 and quantifier elimination. Consider a convergent sequence $(\gamma_m, \sigma_m, \mu_m) \in \mathcal{S}$ with limit (γ, σ, μ). The weak polynomial inequalities (2.2) persist at the limit. For each m, consistency supplies a completely mixed b_m within $1/m$ of σ_m, whose Bayes beliefs at γ_m are within $1/m$ of μ_m.

Factor each prefix reach as

$$R_h(\gamma, b) = c_h(\gamma) a_h(b),$$

where c_h is the product of Nature factors and a_h the product of strategic factors. The finitely many c_h are positive and continuous on Γ. Consequently

$$\rho_m = \max_h \left| \frac{c_h(\gamma)}{c_h(\gamma_m)} - 1 \right| \longrightarrow 0.$$

For $\rho_m < 1$, each reach weight at γ is its weight at γ_m multiplied by a factor in $[1-\rho_m, 1+\rho_m]$. If the old normalized weights are v_h and those factors are t_h, their new normalized weights are $v_h t_h / \sum_j v_j t_j$. Their ℓ^1 difference is at most

$$\sum_h v_h \frac{|t_h - \sum_j v_j t_j|}{\sum_j v_j t_j} \leq \frac{2\rho_m}{1-\rho_m}.$$

This bound is independent of how small strategic reach becomes. Hence the Bayes beliefs of b_m at the fixed γ converge to μ, while $b_m \to \sigma$. Consistency holds at the limit. The graph is closed in the compact parameter/assessment product. □

The proof requires fixed uniformly positive Nature support. It does not close a Bayes graph across a vanishing Nature-support boundary. No such extension is used below.

3 History recovery and the sharp informative fiber

Assume M has full column rank and fix a known left inverse, for example

$$A = (M^\mathsf{T} M)^{-1} M^\mathsf{T}, \qquad AM = I_L. \tag{3.1}$$

Then $q = Ap$. A candidate population vector must satisfy $Mq = p$ and q must be a probability vector; otherwise it is incompatible. For the present informative branch suppose $q_z > 0$ for every z.

Let $Z(h)$ be the terminal descendants of node h, and put

$$r_h = \sum_{z \in Z(h)} q_z.$$

At a Nature edge $h \to ha$, recover

$$c^q_{h,a} = r_{ha}/r_h. \tag{3.2}$$

Perfect recall makes each strategic information set I an antichain. Let $Z(I)$ be the disjoint union of $Z(h)$, $h \in I$, and $Z(I,a)$ the union of descendants of their action-a children. Write $r_I = \sum_{h \in I} r_h$ and $r_{I,a} = \sum_{h \in I} r_{ha}$. The candidate behavior and beliefs are

$$\sigma^q_{I,a} = \frac{r_{I,a}}{r_I}, \qquad \mu^q_{I,h} = \frac{r_h}{r_I}. \tag{3.3}$$

They are positive probability vectors, but behavioral compatibility also requires

$$\frac{r_{ha}}{r_h} = \frac{r_{h'a}}{r_{h'}} \quad \text{for every } h, h' \in I,\ a \in A(I). \tag{3.4}$$

These are testable polynomial equalities after cross multiplication.

Lemma 4 (Recovery and its converse)**.** *A positive q is generated by the stated information structure and an admitted Nature vector exactly when* (3.4) *holds and c^q is admitted. Nature probabilities, behavioral probabilities, and consistent beliefs are then uniquely recovered by* (3.2)–(3.3)*.*

Proof. In any behavioral model, action probabilities agree at all nodes of one information set, and conditional Nature probabilities equal the primitive row. This proves necessity and uniqueness of the ratios. Conversely start with root probability one and use each recovered edge probability. At strategic nodes, (3.4) makes that probability the common $\sigma^q_{I,a}$. Induction along the tree assigns node h probability r_h, because the factor on each edge is the child/parent mass ratio. Thus the generated terminal law is exactly q. Bayes' rule gives μ^q. Every history and action has positive probability, so consistent beliefs are unique; the constant completely mixed strategy sequence supplies a consistency witness. □

The lemma has not assumed equilibrium. In particular, measurement inversion alone does not establish information-set compatibility or the shared Nature restrictions.

For an information set I owned by i, define

$$m^0_{i,I,a}(q) = \frac{\sum_{z\in Z(I,a)} q_z u^0_{i,z}}{r_{I,a}}, \qquad m^{\varphi}_{i,I,a}(q) = \frac{\sum_{z\in Z(I,a)} q_z \varphi_{i,z}}{r_{I,a}}. \tag{3.5}$$

Fix a reference action $a_0(I)$. For each other action form a row

$$D_{i,I,a} = m^{\varphi}_{i,I,a} - m^{\varphi}_{i,I,a_0}, \qquad b_{i,I,a} = m^0_{i,I,a_0} - m^0_{i,I,a}. \tag{3.6}$$

Forced-action information sets add no row.

Lemma 5 (Conditional features and full optimality)**.** *At the recovered compatible profile, the value of choosing a at I and then following the recovered continuation is*

$$Q_{i,I,a}(\beta) = m^0_{i,I,a}(q) + m^{\varphi}_{i,I,a}(q)\cdot\beta = \mathbb{E}_q[u_{i,Z}(\beta) \mid Z(I,a)]. \tag{3.7}$$

Its Bayes assessment is sequentially rational if and only if $D(q)\beta = b(q)$.

Proof. At every hidden node in I, the probability of a is the same positive $\sigma^q_{I,a}$. Conditioning on that action leaves the node distribution equal to μ^q_I. Later play is unchanged, so the terminal conditional distribution is q restricted to $Z(I,a)$. This proves (3.7), with terminal utilities including any already incurred costs under the fixed utility convention.

Sequential rationality implies $Q_{i,I,a} \le \sum_b \sigma^q_{I,b} Q_{i,I,b}$ for every a. All weights are positive, so all action values must coincide. These are precisely the equations $D(q)\beta = b(q)$.

Conversely suppose all those equations hold. Fix an arbitrary full continuation deviation from I, allowing zero action probabilities. At every later own information set J, perfect recall makes the product of earlier own probabilities common to all its nodes. When J is reached it cancels, leaving the original Bayes belief; when it is not reached, it has no payoff contribution. Every node of a later J lies in I's continuation by perfect recall. Replace deviating decisions by the prescribed behavior in decreasing order of remembered own-decision length. At each replacement, later behavior is already prescribed and all local action values coincide by (3.7). The replacement changes no payoff. After finitely many replacements the prescribed continuation remains. Thus no original deviation improves the payoff. Consistency was already established by Lemma 4. □

Theorem 6 (Sharp informative identification)**.** *If the inverse $q = Ap$ is a positive probability vector, failure of $Mq = p$ or* (3.4) *makes the single-assessment model incompatible. Otherwise its entire parameter identified set is*

$$\Gamma_I(p) = \mathcal{F}(q) = \{(\alpha,\beta)\in\Gamma : B\alpha = c^q - c^0,\ D(q)\beta = b(q)\}. \tag{3.8}$$

Every retained parameter has the recovered (σ^q, μ^q) as an actual baseline sequential equilibrium with observation law p.

Proof. Full column rank fixes q for every compatible assessment. Lemma 4 fixes its Nature vector, behavior, and beliefs; Lemma 5 imposes exactly the payoff equations. This proves that every compatible parameter belongs to (3.8). Conversely each point in that polytope supplies the admitted Nature vector and the compatible positive profile, with consistent beliefs and full sequential optimality by the same lemmas. Its terminal and observed laws are q, p. Both inclusions use actual assessments, proving sharpness. □

An empty polytope in (3.8) reports incompatibility with the primitive restrictions. A zero terminal component does not by itself report incompatibility: it places the problem in the general exact branch of Section 9, retaining strategically off-path histories and beliefs.

Let

$$\mathsf{K}(q) = \operatorname{diag}(B, D(q)). \tag{3.9}$$

Full column rank is sufficient for point identification when the fiber is nonempty. In general the fiber is a singleton exactly when all its coordinate minima and maxima coincide; linear programs decide this condition. If a true point lies in the relative interior of Γ, whose affine hull is $E\gamma = f$, the singleton condition is

$$\ker \mathsf{K}(q) \cap \ker E = \{0\}. \tag{3.10}$$

A nonzero vector in this intersection can be moved a small distance in both signs within Γ, preserving the data equations. Conversely, a trivial intersection allows at most one point in their affine-hull intersection. At a boundary, inequalities can make the fiber a singleton without ambient full rank; the coordinate test handles that case.

Observable primitive probabilities need not identify an arbitrary structural index. Null directions of B, additive utility shifts, free scale changes expressible in the family, and payoff changes invisible to continuation differences remain in (3.8) unless joint primitive restrictions remove them.

Recovery uses finite sums, positive divisions, and linear algebra. The feature formulas require terminal-event lists, not exponential enumeration of pure continuation plans. With rational p and rational primitives, the arrays have polynomial bit size in the explicit tree, channel, features, and constraints. Standard rational linear programming decides feasibility and coordinate extrema and returns exact rational witnesses in polynomial bit complexity. For algebraic data its representation must be specified; symbolic population statements keep p as a free variable.

4 Calibration, selection, and a nonempty informative family

Example 7 (An unknown channel is not calibration)**.** For two terminal labels let

$$M_\xi = \begin{pmatrix} (1+\xi)/2 & (1-\xi)/2 \\ (1-\xi)/2 & (1+\xi)/2 \end{pmatrix}, \qquad 0 < \xi \le 1.$$

Each matrix is invertible, but $p_1 = 1/2 + \xi(q_1 - 1/2)$. Unknown ξ permits many pairs (ξ, q_1) with the same observed law. Known full rank without known calibration and aligned labels is insufficient.

Example 8 (Hidden equilibrium mixtures need not be one assessment)**.** In the simultaneous two-player coordination game, both players receive one if their actions L, R match and zero otherwise. Its pure equilibria LL, RR and its independent fair mixed equilibrium are sequential equilibria in the standard perfect-recall tree. Mix their selection probabilities as $(1-\zeta)/2, (1-\zeta)/2, \zeta$, with $0 < \zeta < 1$. The resulting terminal law is positive and satisfies

$$q_{LL} = q_{RR} = (2-\zeta)/4, \qquad q_{LR} = q_{RL} = \zeta/4.$$

The second player's information set joins the nodes after L, R. Its conditional probability of choosing L is $(2-\zeta)/2$ at the first node and $\zeta/2$ at the second. They differ, violating (3.4). No single behavioral assessment in that information structure generates this law. Observed selector strata can instead be conditioned on separately.

Proposition 9 (A primitive-guaranteed correlated Bayesian family)**.** *For every positive integer m, there is a nonempty compact class with unknown correlated private-type law and unknown normalized payoff coefficients, uniquely determined equilibrium behavior, and primitive uniform lower bounds on all terminal probabilities and payoff rank.*

Proof. There are two zero-sum players with private types $t, s \in \{1, \dots, m\}$. Their unknown joint law is a matrix Q with entries summing to one and restrictions

$$\mathsf{Q}_{ts} \geq q_{\min} > 0, \qquad \mathsf{Q}_{tt} \geq 3 \sum_{s \neq t} \mathsf{Q}_{ts}. \tag{4.1}$$

This compact polytope is nonempty when $q_{\min} \leq 1/(4m^2)$, as witnessed by

$$\mathsf{Q} = \frac{1}{m} \left(\frac{3}{4} I_m + \frac{1}{4m} \mathbf{1}\mathbf{1}^{\mathsf{T}} \right).$$

It is invertible: if $\mathsf{Q}x = 0$, take t with $|x_t| = \|x\|_\infty$. Then $\mathsf{Q}_{tt}|x_t| \leq \sum_{s \neq t} \mathsf{Q}_{ts}|x_s| \leq \mathsf{Q}_{tt}|x_t|/3$, forcing $x = 0$.

After learning only their own types, the players simultaneously choose zero or one. Player 1 receives θ_t when both choose one, one when both choose zero, and zero otherwise; player 2 receives the negative. The unknown $\theta_t \in [3/4, 5/4]$ are calibrated by the known zero and one payoffs. Let p_t^a, q_s^a be their probabilities of action one, and put

$$K_{ts}^{\text{type}} = \frac{\mathsf{Q}_{ts}}{\sum_s \mathsf{Q}_{ts}}, \qquad r_t = \frac{1}{1 + \theta_t}.$$

Player 1's typewise indifference and player 2's typewise indifference are respectively

$$K^{\text{type}} q^a = r, \qquad \mathsf{Q}^{\mathsf{T}} \big((1 + \theta) \odot p^a - \mathbf{1} \big) = 0. \tag{4.2}$$

Here $\odot$ means coordinatewise multiplication. Invertibility gives the only possible interior solution

$$p^a = r, \qquad q^a = (K^{\text{type}})^{-1} r. \tag{4.3}$$

It is interior. The diagonal restrictions give $K_{tt}^{\text{type}} \geq 3/4$, so $\|K^{\text{type}} - I\|_\infty \leq 1/2$. The geometric matrix series gives $\|(K^{\text{type}})^{-1}\|_\infty \leq 2$. Since $K^{\text{type}} \mathbf{1} = \mathbf{1}$ and

$$r_t \in [4/9, 4/7], \qquad \|r - \tfrac{1}{2}\mathbf{1}\|_\infty \leq 1/14,$$

we have $\|q^a - \frac{1}{2}\mathbf{1}\|_\infty \leq 1/7$. Both players' action probabilities and their complements exceed $1/3$. Every conditional indifference holds. Each type has one genuine decision, so the positive profile with its Bayes beliefs is a Bayesian Nash and sequential equilibrium.

It is the unique equilibrium behavior. The ex ante game is bilinear and zero-sum on the cubes of typewise probabilities. At the interior opponent in (4.3), all own actions are indifferent. Any other optimal p' must have zero payoff gradient in q^a: otherwise a small feasible perturbation of the interior opponent would lower its guaranteed payoff below the saddle value. The gradient equation in (4.2) and invertibility of Q imply $p' = r$. The symmetric argument with interior p^a gives $K^{\text{type}} q' = r$, hence $q' = q^a$. Every Bayesian equilibrium is root saddle-optimal by the zero-sum minimax inequalities; this argument does not posit general-sum uniqueness.

Every complete terminal (t, s, a, b) consequently has probability at least $q_{\min}/9$. The correlated law Q is recovered by summing complete-history masses. In the payoff fiber, player 1's type-t equation is

$$(K^{\text{type}} q^a)_t \theta_t = 1 - (K^{\text{type}} q^a)_t.$$

Its coefficient matrix in θ is diagonal with entries $r_t \geq 4/9$, proving the claimed rank and conditioning. □

This family may have noisy post-play measurement. On its L terminal labels take, for known rational $0 < \xi \leq 1$,

$$M_\xi = \xi I_L + (1 - \xi) L^{-1} \mathbf{1}\mathbf{1}^{\mathsf{T}}. \tag{4.4}$$

It is stochastic, with eigenvalue one along **1** and ξ on its orthogonal complement. Its Euclidean operator inverse norm is $\|M_{\xi}^{-1}\|_2 = 1/\xi$. It reveals no private type during play. Private types can also be paths revealed in pieces before the terminal actions and persisting through the earlier history. The general theorem itself concerns arbitrary dynamic trees; this family demonstrates nonvacuous primitive conditions.

5 A statistical projection and parameter estimator

For the quantitative informative branch assume $q_z \geq \lambda > 0$. Let m_c be the number of Nature entries, $J = \sum_I (|A(I)| - 1)$ the number of contrast rows, and $R = m_c + J$. Let H_i be the maximum number of player i's decision nodes on a terminal path and $H_* = \max_i H_i$, or zero if there are no strategic decisions. Put $a_M = \|A\|_2$.

Let $\widehat{p}$ be the empirical cell-frequency vector and define

$$\Delta_\lambda = \{v \in \mathbb{R}^L : v_z \geq \lambda,\ \sum_z v_z = 1\}, \quad \widehat{q} = \operatorname*{argmin}_{v \in \Delta_\lambda} \|v - A\widehat{p}\|_2^2, \quad \Delta = \|\widehat{q} - q\|_1. \tag{5.1}$$

The nonempty compact convex domain and strict convexity give a unique projection. If $L\lambda = 1$, it is the constant vector λ. Otherwise its water-filling formula is

$$\widehat{q}_z = \max\{\lambda, (A\widehat{p})_z - \tau\}, \qquad \sum_z \widehat{q}_z = 1.$$

Sort $A\widehat{p}$, enumerate the active-coordinate counts, and solve the last linear equation on each corresponding interval. The feasible interval supplies τ; ties give the same vector. Subtracting τ at free coordinates and truncating the others verifies the projection optimality inequalities. With rational inputs this is an exact rational algorithm.

The statistic $\widehat{q}$ need not satisfy (3.4) or shared Nature restrictions. It is never called an equilibrium law without those checks.

Lemma 10 (Inversion risk and global conditioning)**.** *The projection satisfies*

$$\mathbb{E}\Delta^2 \leq \frac{La_M^2}{n}, \qquad \mathbb{E}\Delta \leq \frac{\sqrt{L}\, a_M}{\sqrt{n}}, \qquad \Delta \leq 2. \tag{5.2}$$

If probability vectors v, w and an event E satisfy $v(E) \geq \lambda$, $w(E) > 0$, and $|f| \leq F_0$, then

$$|\mathbb{E}_v[f \mid E] - \mathbb{E}_w[f \mid E]| \leq \frac{2F_0}{\lambda}\|v - w\|_1. \tag{5.3}$$

For nonnegative vectors x, y with positive sums s, t,

$$\|x/s - y/t\|_1 \leq 2\|x - y\|_1/s. \tag{5.4}$$

Proof. Euclidean projection is nonexpansive. Indeed, if $x = \Pi(v)$ and $y = \Pi(w)$, optimality gives $\langle v - x, y - x\rangle \leq 0$ and $\langle w - y, x - y\rangle \leq 0$. Adding yields

$$\|x - y\|_2^2 \leq \langle x - y, v - w\rangle \leq \|x - y\|_2\|v - w\|_2.$$

Because $q = Ap \in \Delta_\lambda$,

$$\|\widehat{q} - q\|_2 \leq a_M\|\widehat{p} - p\|_2, \qquad \Delta \leq \sqrt{L}\, a_M\|\widehat{p} - p\|_2.$$

The multinomial coordinate variances give $\mathbb{E}\|\widehat{p} - p\|_2^2 = (1 - \|p\|_2^2)/n \leq 1/n$. Cauchy–Schwarz gives the first two assertions of (5.2); the last follows because both vectors are probability vectors.

For (5.3), put $b = v(E)$, $b' = w(E)$, $t_0 = \sum_E v_z f_z$, and $t_0' = \sum_E w_z f_z$. The exact identity

$$\frac{t_0}{b} - \frac{t_0'}{b'} = \frac{t_0 - t_0'}{b} + \frac{t_0'}{b'}\,\frac{b' - b}{b}$$

and $|t_0'/b'| \le F_0$ prove the bound. Only the reference denominator requires a lower bound. Finally

$$x/s - y/t = (x - y)/s + (y/t)(t - s)/s, \qquad |t - s| \le \|x - y\|_1$$

proves (5.4). Both conditional bounds are global; no small-error event is needed. □

Every nonempty prefix or information-action event contains a terminal, so its q- and $\widehat{q}$-mass is at least λ. Define $c^{\mathrm{raw}}_{h,a} = \widehat{q}(Z(ha))/\widehat{q}(Z(h))$ at Nature nodes. Define $\widehat{D}, \widehat{b}$ by (3.5)–(3.6) with $\widehat{q}$ in place of q. They are rational conditional-moment statistics, whether or not $\widehat{q}$ is behaviorally compatible. Lemma 10 gives

$$\begin{aligned} \|c^{\mathrm{raw}} - c^q\|_\infty &\le 2\Delta/\lambda, \\ \max_{j,k} |(\widehat{D} - D)_{j,k}| &\le 4F\Delta/\lambda, \\ \|\widehat{b} - b\|_\infty &\le 4U_0\Delta/\lambda. \end{aligned} \tag{5.5}$$

Uniformly for every β admitted by Γ,

$$\|(\widehat{D} - D)\beta - (\widehat{b} - b)\|_\infty \le 4U\Delta/\lambda. \tag{5.6}$$

This follows from the entry bounds and $\|\beta\|_1 \le B_1$, or by applying (5.3) directly to $u_{i,Z}(\beta)$.

Put $C = \max\{2, 4U\}/\lambda$. The estimator solves the rational LP

$$\begin{aligned} &\text{minimize } t \\ &\quad \text{over } (\alpha, \beta) \in \Gamma, \quad t \ge 0 \\ &\text{subject to } \|B\alpha - (c^{\mathrm{raw}} - c^0)\|_\infty \le t, \qquad \|\widehat{D}\beta - \widehat{b}\|_\infty \le t. \end{aligned} \tag{5.7}$$

Its optimizer $(\widehat{\gamma}, \widehat{t})$ exists because the minimized residual is continuous on compact Γ. A deterministic exact rational choice is obtained by optimizing t and then successively minimizing coordinates on its optimizer face.

Theorem 11 (Every-sample fit and conditioned risk)**.** *For every point γ of the entire true fiber $\mathcal{F}(q)$,*

$$\widehat{t} \le C\Delta, \tag{5.8}$$

$$\|\mathsf{K}(q)(\widehat{\gamma} - \gamma)\|_\infty \le 2C\Delta. \tag{5.9}$$

If $\mathsf{K}(q)$ has full column rank with smallest singular value $\kappa > 0$, the fiber is a singleton γ_0, and for any known $G \ge \mathrm{diam}_2(\Gamma)$,

$$\mathbb{E}\|\widehat{\gamma} - \gamma_0\|_2^2 \le \min\left\{G^2, \frac{4RC^2La_M^2}{\kappa^2 n}\right\}. \tag{5.10}$$

Proof. Every fiber point has residual at most $C\Delta$ in (5.7), proving (5.8). The fitted chance block differs from the true chance block by at most $\widehat{t} + 2\Delta/\lambda$. The fitted payoff block satisfies $\|D\widehat{\beta} - b\|_\infty \le \widehat{t} + 4U\Delta/\lambda$, since $\widehat{\gamma} \in \Gamma$ permits (5.6). Both are at most $2C\Delta$. Subtracting any fiber point's exact equations proves (5.9). Under full rank, $\|\widehat{\gamma} - \gamma_0\|_2 \le 2\sqrt{R}\,C\Delta/\kappa$. Square, use (5.2), and separately use both points' membership in Γ to obtain (5.10). □

The bound is unconditional, without an estimated inverse, a logarithm of n, or a fixed-probability remainder. A valid G is the square root of the sum of squared coordinate range widths of Γ, found by $2d$ LPs. Computing its exact Euclidean diameter is not claimed to be an LP. Under rank deficiency, (5.9) estimates the observable linear combinations; it does not identify an arbitrary chosen point of the fiber.

6 Whole-fiber confidence and diameter certificates

Fix a confidence error $0 < \eta < 1$, distinct from the chance parameter α. The elementary coordinate Hoeffding bound gives, with probability at least $1 - \eta$,

$$\|\widehat{p} - p\|_\infty \le h_n, \qquad h_n = \sqrt{\frac{\log(2K/\eta)}{2n}}.$$

For completeness, the logarithm of the centered moment-generating function of a Bernoulli variable has value and derivative zero at zero, and second derivative equal to a tilted Bernoulli variance, at most $1/4$. Twice integrating yields $\log \mathbb{E}e^{t(X-\mathbb{E}X)} \le t^2/8$ for every real t. Independence and exponential Markov, optimized at $t = 4h$, give each upper tail e^{-2nh^2}; negative t gives the same lower-tail bound. A union bound over both tails and all K coordinates proves the assertion.

Define

$$e_n = \min\{2, \sqrt{LK}\, a_M h_n\}. \tag{6.1}$$

The same event gives $\Delta \le e_n$. Our outer confidence polytope is

$$\begin{aligned} \mathcal{C}_n = \{(\alpha, \beta) \in \Gamma : \|B\alpha - (c^{\mathrm{raw}} - c^0)\|_\infty \le 2e_n/\lambda, \\ \|\widehat{D}\beta - \widehat{b}\|_\infty \le 4Ue_n/\lambda\}. \end{aligned} \tag{6.2}$$

Theorem 12 (Finite-sample coverage of the entire fiber)**.** *Uniformly over the declared quantitative law class,*

$$\mathbb{P}\{\mathcal{F}(q) \subseteq \mathcal{C}_n\} \ge 1 - \eta. \tag{6.3}$$

This remains valid at rank-deficient chance/payoff operators.

Proof. On $\Delta \le e_n$, (5.5)–(5.6) imply both inequalities in (6.2) simultaneously for every point of the exact fiber. The implication is deterministic on the single concentration event; no parameter or equilibrium is selected. □

A fully rational implementation can replace e_n everywhere by a conservative rational radius. For rational η, let b_n be the least nonnegative integer with $2^{b_n} \ge 2K/\eta$, and put

$$\bar{h}_n = \frac{\lceil \sqrt{2nb_n} \rceil}{2n}, \qquad g = \lceil \sqrt{LK} \rceil, \qquad \bar{e}_n = \min\{2, g\bar{a}_M \bar{h}_n\}, \tag{6.4}$$

where rational $\bar{a}_M \ge a_M$ can upper-bound the Frobenius norm of A. Since $\log(2K/\eta) \le b_n$ and $\bar{h}_n \ge \sqrt{b_n/(2n)}$, we have $\bar{e}_n \ge e_n$. Integer comparisons and square roots construct it. At fixed input and confidence it is $O(n^{-1/2})$, with an additional $O(n^{-1})$ rounding term. All subsequent formulas permit this replacement. The radius cap uses the deterministic $\Delta \le 2$.

The polytope can be empty outside the coverage event and is reported as empty. An optional replacement by Γ is conservative but is not used in the expected-diameter proof. At radius two, $\mathcal{C}_n = \Gamma$: the compared chance entries are probabilities and every payoff contrast has magnitude at most $2U$.

For a nonempty $\mathcal{C}_n$, obtain each coordinate's minimum l_j and maximum u_j with $2d$ LPs. Then

$$\mathrm{diam}_2(\mathcal{C}_n) \le \sqrt{\sum_{j=1}^{d} (u_j - l_j)^2}.$$

With the rational radius these endpoints are exact rational extrema. The coordinate ranges are sharp for this outer polytope; the resulting bounding-box diameter need not equal its Euclidean diameter.

Let $\widehat{\mathsf{K}} = \operatorname{diag}(B, \widehat{D})$. If its smallest singular value $\widehat{\kappa}$ is positive, the difference of any two candidate vectors obeys the rowwise tolerances in (6.2), giving

$$\begin{aligned}\operatorname{diam}_2(\mathcal{C}_n) &\le \min\left\{G, \frac{2e_n\sqrt{m_c(2/\lambda)^2 + J(4U/\lambda)^2}}{\widehat{\kappa}}\right\} \\ &\le \min\{G, 2\sqrt{R}\,Ce_n/\widehat{\kappa}\}.\end{aligned} \tag{6.5}$$

A certified positive lower bound for $\widehat{\kappa}$ suffices. An estimated rank test is unnecessary for coverage and does not, by itself, establish population point identification.

Proposition 13 (Every-sample and expected diameter)**.** *Suppose the true* $\mathsf{K}(q)$ *has smallest singular value* $\kappa > 0$*. With* $\operatorname{diam}(\varnothing) = 0$*,*

$$\operatorname{diam}_2(\mathcal{C}_n) \le \min\{G, 2\sqrt{R}\,C(e_n + \Delta)/\kappa\}, \tag{6.6}$$

$$\mathbb{E}\operatorname{diam}_2(\mathcal{C}_n) \le \min\{G, 2\sqrt{R}\,C(e_n + \sqrt{L}\,a_M/\sqrt{n})/\kappa\}. \tag{6.7}$$

Proof. Fix any true fiber point γ_0. For any $\gamma = (\alpha, \beta) \in \mathcal{C}_n$, the chance block differs from the true one by at most $2(e_n + \Delta)/\lambda$. For its payoff block, (5.6) applies to this candidate $\beta \in \Gamma$, giving a difference at most $4U(e_n + \Delta)/\lambda$. Thus

$$\|\mathsf{K}(q)(\gamma - \gamma_0)\|_\infty \le C(e_n + \Delta).$$

Every candidate is within $\sqrt{R}\,C(e_n + \Delta)/\kappa$ of γ_0, whether or not γ_0 belongs to this sample's confidence polytope. Triangle inequality proves (6.6); the empty set satisfies it by convention. Take expectations, use (5.2), and retain the independent G cap to obtain (6.7). □

There is no ηG remainder. Fixed positive conditioning gives an $O(n^{-1/2})$ expected diameter at fixed confidence. No point- shrinking claim is made when a nontrivial true fiber remains. For reference, (5.5) also implies

$$\|\widehat{\mathsf{K}} - \mathsf{K}(q)\|_2 \le (4F\sqrt{Jd_\beta}/\lambda)\Delta.$$

This Frobenius bound can transfer a supplied population rank margin on a concentration event; neither the estimator risk nor (6.6) requires such a transfer.

7 An actual assessment and every-continuation certificates

Use one behavior vector per information set,

$$\widehat{\sigma}_{I,a} = \frac{\widehat{q}(Z(I,a))}{\widehat{q}(Z(I))}, \qquad c^\star = c^0 + B\widehat{\alpha}, \qquad u^\star_{i,z} = u_{i,z}(\widehat{\beta}). \tag{7.1}$$

The action events partition $Z(I)$, so this is a proper distribution. Its numerator is at least λ and denominator at most one; hence $\widehat{\sigma}_{I,a} \ge \lambda$. Let $q^\star, r^\star_h$ be the actual terminal and prefix probabilities obtained by multiplying $c^\star, \widehat{\sigma}$ along the original tree, and define

$$\mu^\star_{I,h} = \frac{r^\star_h}{\sum_{h' \in I} r^\star_{h'}}. \tag{7.2}$$

Because $\widehat{\gamma} \in \Gamma$, all shared Nature restrictions are exactly satisfied. Let $c_{\min} > 0$ be the minimum included Nature entry over Γ, computed by finitely many LPs; take $c_{\min} = 1$ if there are no Nature edges. Every terminal has probability at least $\min\{c_{\min}, \lambda\}^H > 0$. Thus $\mu^\star$ is defined and satisfies Bayes' rule exactly. The constant completely mixed strategy sequence proves consistency. In general $q^\star \ne \widehat{q}$; node-normalized $\widehat{q}$ weights are not used as these beliefs.

Equations (5.5) and (5.8) give

$$\|c^\star - c^q\|_\infty \le \widehat{t} + 2\Delta/\lambda \le 2C\Delta, \qquad \|\widehat{\sigma} - \sigma^q\|_\infty \le 2\Delta/\lambda \le C\Delta.$$

Every edge factor therefore differs by at most $\delta_{\text{edge}} = 2C\Delta$. For factors in $[0,1]$, telescoping products gives

$$\left|\prod_{j=1}^{s} x_j - \prod_{j=1}^{s} y_j\right| \le \sum_{j=1}^{s} |x_j - y_j|.$$

Each path has at most H factors. Consequently

$$|q_z^\star - q_z| \le H\delta_{\text{edge}}, \qquad \|q^\star - q\|_1 \le 2LHC\Delta. \tag{7.3}$$

The last right-hand side may always be capped by two. The same prefix-product bound and (5.4), using true information-set reach at least λ, yield

$$\|\mu_I^\star - \mu_I^q\|_1 \le \frac{2|I|H\delta_{\text{edge}}}{\lambda} \le \frac{4N_{\text{tr}}HC\Delta}{\lambda}. \tag{7.4}$$

This is a global normalization bound, requiring no small-error split or estimated reach lower bound.

All following action values concern this *actual* reconstructed game. The common positive action probability at I cancels in conditioning, so

$$Q_{i,I,a}^\star = \mathbb{E}_{q^\star}[u_{i,Z}(\widehat{\beta}) \mid Z(I,a)], \qquad V_{i,I}^\star = \sum_a \widehat{\sigma}_{I,a} Q_{i,I,a}^\star. \tag{7.5}$$

Define the exact local certificate

$$\ell_{\text{loc}} = \max_{i,I,a} [Q_{i,I,a}^\star - V_{i,I}^\star]_+,$$

or zero if no player moves. It is computable as a rational number under the rational implementation.

The true-q contrasts at $\widehat{\beta}$ are the entries of $D(q)\widehat{\beta} - b(q)$, bounded by $2C\Delta$. Apply (5.3) with reference q, other law $q^\star$, and payoff bounded by U, separately for two actions. With (7.3) this gives

$$|Q_{i,I,a}^\star - Q_{i,I,a_0}^\star| \le C_{\text{loc}}\Delta, \qquad C_{\text{loc}} = 2C + \frac{8ULHC}{\lambda}. \tag{7.6}$$

Two arbitrary actions differ by at most $2C_{\text{loc}}\Delta$; the prescribed value is a convex combination, whence

$$\ell_{\text{loc}} \le 2C_{\text{loc}}\Delta. \tag{7.7}$$

Lemma 14 (Quantitative backward replacement)**.** *In any finite perfect-recall game with positive Nature probabilities, completely mixed behavior, and its exact Bayes beliefs, let ℓ_{loc} be the local certificate just defined using that game. For every player i, starting information set I, and full continuation deviation δ_i,*

$$V_{i,I}(\delta_i) - V_{i,I}(\textit{prescribed continuation}) \le H_i \ell_{\text{loc}}. \tag{7.8}$$

The deviation can randomize, use zero action probabilities, and depend on the full information histories allowed by the tree.

Proof. At an own information set J, perfect recall fixes the remembered sequence of own earlier information sets and actions. The product of one's earlier behavioral probabilities is common at all nodes of J. Replacing that earlier behavior multiplies their reach weights by the same factor. A positive factor cancels in conditional beliefs; a zero factor makes J unreachable. Thus any reached J under a deviation has the original Bayes belief. The same argument applies to play started conditionally at I: a later own J remembers I and the action there, so all its nodes belong to that continuation, and the common factor cancels.

Order own information sets after I, including I, by decreasing remembered own-decision length. Replace deviating choices by the prescribed ones in this order. When replacing J, all later own choices are prescribed. If J is reached, its belief is the one used to compute $Q_{i,J,a}$. The current deviating mixture can exceed the prescribed mixture value by at most ℓ_{loc}. The reduction in payoff from replacing it is therefore at most ℓ_{loc} times the probability of reaching J from I. An unreached set contributes zero.

Earlier own decisions are still the original deviation's decisions. Replacements later than J cannot affect its reach, and sets with the same remembered depth cannot follow one another. All these reach probabilities are therefore those under the *same original* deviation from I. Their sum is its expected number of own decision nodes visited, at most H_i. Summing the payoff changes telescopically proves (7.8). This proves the full continuation bound rather than assuming a one-step-deviation principle.

Behavioral deviations are covered directly. For privately randomized complete plans, condition on the private randomization, apply the bound to each deterministic plan, and average. □

Let $\mathrm{Reg}_{\mathrm{SE}}$ be the maximum positive full continuation improvement over all players and information sets, evaluated with $\mu^\star$ in the actual reconstructed game. Then

$$\begin{aligned} \mathrm{Reg}_{\mathrm{SE}}(\widehat{\gamma},\widehat{\sigma},\mu^\star) &\le H_*\ell_{\mathrm{loc}} \le C_{\mathrm{eq}}\Delta, \\ C_{\mathrm{eq}} &= 2H_*C_{\mathrm{loc}} = 4H_*C(1+4ULH/\lambda). \end{aligned} \tag{7.9}$$

Equation (5.2) and the concentration event give

$$\mathbb{E}\,\mathrm{Reg}_{\mathrm{SE}}^2 \le \min\{4U^2, C_{\mathrm{eq}}^2 L a_M^2/n\}, \tag{7.10}$$

$$\mathbb{P}\{\mathrm{Reg}_{\mathrm{SE}} \le \min\{2U, C_{\mathrm{eq}}e_n\}\} \ge 1-\eta. \tag{7.11}$$

Both utility comparisons lie in $[-U,U]$, which justifies the caps. The rational radius can replace e_n.

More strongly, $H_*\ell_{\mathrm{loc}}$ is an a posteriori certificate valid on every data set without knowledge of q, Δ, or the true parameter. Primitive feasibility, common information-set behavior, and Bayes consistency are exact. Only sequential optimality has a residual. If $\ell_{\mathrm{loc}} = 0$, the assessment is an exact sequential equilibrium; otherwise that assertion is not made.

8 Effective statistical computation and its limits

For rationally encoded $M, c^0, B, u^0, \varphi, \lambda, B_1, \Gamma, \eta$, the following procedure is finite and explicit:

1. Compute A by exact rational linear algebra, form the cell frequencies, and compute $\widehat{q}$ by sorting and water filling.
2. Sum prefix and information-action masses and features, then form $c^{\mathrm{raw}}, \widehat{D}, \widehat{b}$.
3. Solve (5.7) over the full joint Γ; construct $\bar{e}_n$ and $\mathcal{C}_n$.

4. Decide confidence-polytope feasibility and, when nonempty, compute its coordinate extrema and certified diameter bound.

5. Form $c^\star, \widehat{\sigma}, q^\star, \mu^\star$, compute actual action values, and report the certificate $H_* \ell_{\mathrm{loc}}$.

There are polynomially many rational operations and LPs in the explicit tree, channel, utility features, and parameter inequalities. Sorting uses $O(L \log L)$ comparisons. Even naively listing the terminal incidence of each event has polynomial explicit-input size. Empirical event denominators are at least λ. Actual Bayes denominators are positive, possibly as small as $\min\{c_{\min}, \lambda\}^H$; finite path products and sums still have polynomial rational bit length in the explicit tree and input bit length. Small probability is not large representation length by itself.

Standard exact rational linear algebra has polynomial bit complexity. We use the classical polynomial-bit algorithm for rational linear programming, with explicitly listed rational constraints of polynomial bit length. Deterministic optimizer selection takes at most d additional LPs. Integer comparisons and square roots construct (6.4); a rational sum of squares supplies a certified upper bound for $\|A\|_{\mathrm{F}}^2$. Spectral computation is not needed for coverage, the estimator, or the coordinate-range certificate. To report a spectral diameter bound, a rational positive-semidefiniteness certificate for $\widehat{\mathsf{K}}^{\mathsf{T}}\widehat{\mathsf{K}} - \kappa_{\mathrm{low}}^2 I_d$ supplies a valid positive κ_{low}.

The procedure is polynomial in the explicit input and in $\log n$ when cell counts are supplied. Reading n raw observations costs at least n operations. This is neither a strongly polynomial LP claim nor a polynomial claim in succinct horizon T. Algebraic data need their exact representation and corresponding arithmetic guarantees; arbitrary unencoded real input is excluded from finite-bit assertions.

The population fiber is sharp. Its confidence polytope is a conservative outer set, not another sharp population fiber. Finite-sample incompatibility of the raw projection with (3.4) is expected, not an exact rejection test. An empty confidence polytope, in contrast, has probability at most η under the declared model because the nonempty true fiber is contained on the coverage event.

The quantitative constants expose measurement conditioning a_M, history conditioning λ, payoff bounds, and the identification margin κ. They do not remain bounded as those conditions degenerate. No optimal dependence on these constants is claimed. Unknown calibration or a rank-deficient channel does not inherit the inversion rate. Without full history support the generic exact method remains valid, but these ratio rates require another analysis. A nontrivial payoff/chance fiber remains set-valued.

9 Sharp general policy sets and comparative selection

Return to the generic class, permitting rank-deficient M and strategically zero terminal probabilities. Let $\mathcal{S}_0$ be the baseline equilibrium graph and $\mathcal{S}_\ell$ the policy graph from Section 2. Let $O(\gamma, \sigma) = Mq(\gamma, \sigma)$. For population law p, define

$$\begin{aligned} \mathcal{A}(p) &= \{(\gamma, e^0) \in \mathcal{S}_0 : O(\gamma, \sigma^0) = p\}, \\ \Gamma_I(p) &= \mathrm{proj}_\gamma \mathcal{A}(p), \\ \mathcal{J}_\ell(p) &= \{(\gamma, e^0, e^\ell) : (\gamma, e^0) \in \mathcal{A}(p),\ (\gamma, e^\ell) \in \mathcal{S}_\ell\}, \\ \mathcal{W}_I(\ell; p) &= \{W_\ell(\gamma, \sigma^\ell) : (\gamma, e^0, e^\ell) \in \mathcal{J}_\ell(p)\}. \end{aligned} \tag{9.1}$$

Theorem 15 (Exact ranges with attained scalar endpoints)**.** *These are exactly the canonical parameter set and all-selection policy range. They are compact semialgebraic sets, possibly empty. If $\Gamma_I(p) \neq \varnothing$, each policy range is nonempty and its scalar minimum and maximum are attained. The full range need not be an interval.*

Proof. Lemma 1 makes every feasible tuple an actual baseline SE producing p and an actual policy SE at the same parameter. Conversely every canonical compatible model supplies such a tuple. This proves sharpness in both directions. Lemma 3 and the closed observation equality make $\mathcal{A}(p)$ compact. Its projection and the shared-parameter product defining $\mathcal{J}_\ell(p)$ are compact. Continuity of (1.4) preserves compactness of its image; polynomial quantification preserves semialgebraicity. Nonempty policy fibers follow from Lemma 2, and continuous scalar functions attain extrema on compact sets. None of these operations fills gaps in a disconnected image. □

Without a supplied cross-policy selector, a sharp same-parameter adverse comparison range is

$$\begin{aligned}\{W_\ell(\gamma,\sigma^\ell) - W_{\ell'}(\gamma,\sigma^{\ell'}) : (\gamma,e^0)\in\mathcal{A}(p),\\ (\gamma,e^\ell)\in\mathcal{S}_\ell,\ (\gamma,e^{\ell'})\in\mathcal{S}_{\ell'}\}.\end{aligned} \tag{9.2}$$

It permits every pair of policy equilibria, retaining the same structural parameter. Its endpoints are attained by the same argument. Subtracting marginal endpoints can use different parameters and is only an explicitly broader bound. A supplied closed semialgebraic joint selection relation can instead be intersected with (9.2); nonemptiness must then be checked. No such relation is assumed by default.

10 Exact computation, endpoint errors, and coverage transfer

All defining constraints above are explicit finite first-order formulas. Classical effective real-closed-field quantifier elimination [2] gives descriptions of the full Γ_I and $\mathcal{W}_I$, including disconnected parts. Emptiness is checked first. Rational and represented algebraic numerical inputs are admissible; alternatively p is left symbolic.

For a nonempty feasible tuple set $\mathcal{J}$, its scalar extrema are specified by

$$\begin{aligned}&\exists y\in\mathcal{J}:\ W(y)=w_{\min}, \qquad &&\forall y\in\mathcal{J}:\ W(y)\ge w_{\min},\\ &\exists y\in\mathcal{J}:\ W(y)=w_{\max}, \qquad &&\forall y\in\mathcal{J}:\ W(y)\le w_{\max}.\end{aligned} \tag{10.1}$$

Quantifier elimination and real-root isolation compute these algebraic numbers. Semialgebraic sample-point extraction supplies algebraic tuples attaining them, including actual baseline and policy SE assessments. Their incentive residuals and consistency constraints are exact.

A rational bound $|W|\le B_W$ is computable. First bound the coordinates of compact Γ by rational LPs; then bound each terminal polynomial by the sum of its absolute coefficient-monomial bounds. Expected W is a convex combination of those terminal values. Start from rational $[L_{\text{init}},U_{\text{init}}]$, for instance $[-B_W,B_W]$, and any feasible tuple. For the maximum, query whether there exists a feasible tuple with $W\ge q_{\text{mid}}$. A true answer gives a lower bound and a witness; a false answer gives an upper bound. For the minimum use $W\le q_{\text{mid}}$, with reversed interval updates. After

$$\max\left\{0,\left\lceil\log_2\frac{U_{\text{init}}-L_{\text{init}}}{\varepsilon}\right\rceil\right\} \tag{10.2}$$

queries when the initial width is positive, the interval width is at most ε. Zero width requires no queries. For the maximum, the saved witness has value at least the maintained lower bound, even if no midpoint query was true because the initial witness is at least L_{init}. It is within ε of the global maximum. The minimum argument is the reverse. The certificate records the formulas, initial bounds, true-query witnesses, and exact real-algebraic decision certificates for false queries. This is a complete algorithm.

Unrestricted quantifier elimination may be expensive. When *all* compiled variable counts, degrees, and structural tree bounds are fixed, the operation bound and integer bit-size clause of

Basu's Theorem 2.27 [2, p. 16] give polynomial bit cost in dense coefficient/formula size and rational precision-input length. Theorem 2.4 [2, p. 6] gives the usual CAD arithmetic bound. All belief, tremble, continuation, and primitive variables must be counted. This claim is distinct from polynomial explicit-tree LP computation and implies no polynomial succinct-horizon algorithm.

For generic iid sampling, with probability at least $1-\eta$, $\|\widehat{p}-p\|_\infty \le h_n$, by the Bernoulli argument of Section 6. Replace the baseline observation equality by

$$\|O(\gamma,\sigma^0)-\widehat{p}\|_\infty \le h_n,$$

or a rational conservative radius. On that event, every tuple in $\mathcal{A}(p)$ remains feasible with its original exact SE witness. Projection covers the whole parameter set; attaching every exact policy equilibrium covers every policy range, simultaneously on the same event. The same reasoning covers (9.2). This generic coverage argument is retained from [4].

More generally, let E_n be a measurable event of probability at least $1-\eta$ on which an outer parameter set $\mathcal{C}_n$ contains the whole identified set. Define

$$\mathcal{V}_n(\ell)=\{W_\ell(\gamma,\sigma^\ell):\gamma\in\mathcal{C}_n\cap\Gamma,\ (\gamma,e^\ell)\in\mathcal{S}_\ell\}.$$

Set inclusion gives the following common-event assertion, writing ω for the sampled data:

$$\mathbb{P}(E_n)\ge 1-\eta,\qquad \forall\omega\in E_n\ \forall\ell:\quad \mathcal{W}_I(\ell;p)\subseteq\mathcal{V}_n(\ell;\omega). \tag{10.3}$$

It holds for every policy on one measurable event, without needing measurability of an uncountable intersection of policy-specific events. Thus the informative confidence polytope can be used without solving the baseline equilibrium problem again. Use the identical joint selection relation on both sides if one is part of the target. Exact policy ranges over the outer polytope can be computed by the preceding algorithm. Certified outer supersets preserve coverage; a small equilibrium residual alone does not imply outer containment. The approximate baseline assessment in Section 7 is never substituted for an exact policy equilibrium here.

11 Zero-sum policies: realization flows and a value LP

For the faster policy branch there are two strategic players and Nature, perfect recall, and $u_{2,z}=-u_{1,z}$. The target W_ℓ is precisely player 1's expected zero-sum utility, not an arbitrary action, allocation, revenue, or welfare statistic. The following classical sequence construction [3, Section 2, pp. 249–250] is proved to specify its value and certificates.

Give an action its information-set-and-action label. A player's sequence is its ordered list of own action labels along a history. Perfect recall gives a single preceding sequence $s(I)$ at all nodes in one information set I. For player i, the sequence set consists of the empty sequence and each $s(I)a$. Its realization vector x_i obeys

$$x_i(\varnothing)=1,\qquad x_i(s)\ge 0,\qquad \sum_{a\in A(I)} x_i(s(I)a)=x_i(s(I))\quad\text{for every } I. \tag{11.1}$$

Write these equalities as $E_ix_i=e_i$, where E_i has entries in $\{-1,0,1\}$ and $e_i=(1,0,\dots,0)$. Let $\mathcal{R}_i$ be this nonnegative flow polytope.

Lemma 16 (Behavior and realization weights)**.** *The nonempty compact polytope $\mathcal{R}_i$ describes exactly the own terminal realization weights of behavioral strategies, against every opponent strategy.*

Proof. A behavior assigns each sequence the product of its own probabilities, which satisfies (11.1). Conversely, from a feasible vector set

$$\sigma_i(I, a) = x_i(s(I)a)/x_i(s(I))$$

when the denominator is positive, using any distribution otherwise. If a parent weight is zero, nonnegativity and the flow equation force every child weight to zero. Induction on sequence length therefore recovers every prescribed realization weight, including zero-prefix sequences. Each coordinate lies between zero and its parent, hence in $[0, 1]$. The constraints are closed and admit uniform behavior, proving compactness and nonemptiness. For any opponent behavior, a terminal probability depends on player i only through its final own sequence product. Thus the equivalence is opponent-independent. □

Preselected random pure plans are covered as well: their probabilities of agreeing with each sequence satisfy the same flow equations, and the behavioral reconstruction has the same terminal law against every opponent. Exponential normal-form enumeration is unnecessary.

For a policy terminal z, let $s_i(z)$ be player i's final sequence and let $k_z(\alpha)$ be the product of Nature probabilities on its complete path. Define

$$G_{s,t}(\gamma) = \sum_{\substack{z: s_1(z)=s \\ s_2(z)=t}} k_z(\alpha) u_{1,z}(\beta). \tag{11.2}$$

For $x \in \mathcal{R}_1, y \in \mathcal{R}_2$, expected utility is

$$x^{\mathsf{T}} G(\gamma) y. \tag{11.3}$$

Indeed each terminal probability is $k_z(\alpha)x(s_1(z))y(s_2(z))$; summing its utility proves the identity. There are $1+\sum_{I \text{ of } i} |A(I)|$ variables and $1+\#\{I \text{ of } i\}$ rows for player i, linear in the explicit tree size. Each leaf contributes to one matrix location. Leaves sharing a sequence pair are summed, so a sparse matrix has at most the number of leaves nonzero locations.

For fixed x, player 2's minimization is

$$\min_{y \geq 0,\ E_2 y = e_2} (G^{\mathsf{T}} x)^{\mathsf{T}} y.$$

It is feasible and bounded. LP strong duality identifies its value with $\max_v e_2^{\mathsf{T}} v$ subject to $E_2^{\mathsf{T}} v \leq G^{\mathsf{T}} x$, where v is free. Maximizing also over x gives the primal-dual pair

$$\begin{aligned}
&\text{(P)} \quad \max_{x,v} e_2^{\mathsf{T}} v \\
&\qquad \text{subject to } E_1 x = e_1,\ x \geq 0, \quad E_2^{\mathsf{T}} v \leq G^{\mathsf{T}} x,\ v \text{ free;} \\
&\text{(D)} \quad \min_{y,w} e_1^{\mathsf{T}} w \\
&\qquad \text{subject to } E_2 y = e_2,\ y \geq 0, \quad E_1^{\mathsf{T}} w \geq G y,\ w \text{ free.}
\end{aligned} \tag{11.4}$$

The signs can be checked directly: every feasible pair satisfies

$$e_2^{\mathsf{T}} v = y^{\mathsf{T}} E_2^{\mathsf{T}} v \leq x^{\mathsf{T}} G y \leq x^{\mathsf{T}} E_1^{\mathsf{T}} w = e_1^{\mathsf{T}} w. \tag{11.5}$$

Equality constraints give free dual variables, and nonnegative realization variables give the displayed inequality directions. The direct dual construction of (P) gives (D).

Both LPs are feasible: choose any x, respectively y, and use a dual optimizer for the bounded opponent best-response LP. The primal objective is bounded above and the dual objective below

by the finite expected-utility range. Strong duality yields equal attained optima $v_\ell(\gamma)$, and proves the minimax identity

$$v_\ell(\gamma) = \max_{x\in\mathcal{R}_1}\min_{y\in\mathcal{R}_2} x^{\mathsf{T}}Gy = \min_{y\in\mathcal{R}_2}\max_{x\in\mathcal{R}_1} x^{\mathsf{T}}Gy. \tag{11.6}$$

At equality in (11.5), x guarantees at least v_ℓ against every y', and y at most v_ℓ against every x'. They form a saddle pair. Conversely every root Nash payoff equals this value.

Exact rational vectors satisfying (11.4) certify

$$\underline{v}_\ell = e_2^{\mathsf{T}}v \le v_\ell(\gamma) \le e_1^{\mathsf{T}}w = \overline{v}_\ell. \tag{11.7}$$

Checking rows, signs, and inner products gives a finite exact certificate. A gap at most ε_ℓ certifies that numerical error for the *value*; zero gap gives its exact value.

For rational γ, the matrix entries are computed from path products and leaf sums with polynomial bit length and time in the explicit tree and input. Rational LP has polynomial-bit optimal primal and dual certificates and a polynomial-bit algorithm. Thus the exact zero-sum value has polynomial explicit-tree computation, without fixing structural dimension. This is not a strongly polynomial LP assertion. The identities hold for real γ, but unencoded real numerical input has no asserted bit algorithm.

An arbitrary LP saddle behavior, especially behavior freely completed at zero-realization information sets, is *not* a certified sequential assessment off path. The LP neither constructs consistent beliefs nor imposes off-path sequential rationality.

Lemma 17 (The LP computes the value of every SE)**.** *Every sequential equilibrium of a zero-sum policy game has expected player-1 utility $v_\ell(\gamma)$, and at least one exists. Therefore, when W_ℓ is that utility,*

$$\mathcal{W}_I(\ell; p) = \{v_\ell(\gamma) : \gamma \in \Gamma_I(p)\}. \tag{11.8}$$

Proof. A sequential equilibrium is root Nash. To prove this directly, partition each player's initially reachable paths by its first own information set. The probabilities of reaching those sets depend only on Nature and opponents, not on its deviation. At a positive-reach first set, consistency gives the actual Bayesian conditional belief, and (2.2) compares the entire alternative continuation. Multiply by those reach probabilities and add to prove the root Nash inequality. Paths on which the player never moves have unchanged payoff; a zero-reach first set remains unreachable under its deviation.

The root behavior gives feasible realizations by Lemma 16. The saddle value (11.6) forces its zero-sum utility to equal v_ℓ. Existence follows from Lemma 2. Taking all compatible parameters proves (11.8). □

12 Value stability without equilibrium-selector continuity

Fix one policy tree and two parameters $\gamma, \gamma' \in \Gamma$. Let H_ℓ^{N} be the maximum number of Nature draws along a terminal path, including initial states, signals, and transitions. Define

$$\begin{aligned} \delta_c &= \max_{\text{Nature nodes } h} \sum_{e \text{ leaving } h} |c_e(\alpha) - c_e(\alpha')|, \\ \delta_u &= \max_z |u_{1,z}(\beta) - u_{1,z}(\beta')|, \qquad U_\ell = \max_{\gamma\in\Gamma,z} |u_{1,z}(\beta)|. \end{aligned} \tag{12.1}$$

The maximum over no Nature nodes is zero; rational LPs compute U_ℓ exactly.

Lemma 18 (Uniform fixed-profile perturbation)**.** *For every pair of behavioral strategies, held fixed as functions on the original information sets, the two expected utilities differ by at most*

$$\delta_u + U_\ell H_\ell^{\mathrm{N}} \delta_c. \tag{12.2}$$

Proof. Couple the plays while their histories agree, using the same strategic random draws at identical information sets. At each Nature node use a maximal coupling of its two finite rows, whose failure probability is their total variation and at most $\delta_c/2$. There are at most H_ℓ^{N} opportunities for a first failure. Summing conditional first-failure probabilities bounds the probability of different terminal histories by $H_\ell^{\mathrm{N}}\delta_c/2$. At identical terminals the utility difference is at most δ_u. At different terminals, separate the parameter change at one terminal, at most δ_u, from the difference between two utilities at γ', at most $2U_\ell$. Taking expectation proves (12.2). Private persistent information is preserved: before histories differ, the same history uses the same information-set distribution. □

If two functions differ uniformly by at most ε, their minima and maxima differ by at most ε: the pointwise inequalities imply both one-sided bounds after optimization. Apply this fact twice to (11.6) and Lemma 18 to get

$$|v_\ell(\gamma) - v_\ell(\gamma')| \le \delta_u + U_\ell H_\ell^{\mathrm{N}} \delta_c. \tag{12.3}$$

No unique equilibrium, strict incentive gap, nonsingular policy Jacobian, or continuous selector is used. For Nature coefficient rows B_e, set

$$\begin{aligned} L_{c,\ell} &= \max_h \sum_{e \text{ leaving } h} \|B_e\|_1, \\ L_{u,\ell} &= \max_z \|\varphi_{1,z}\|_1, \qquad L_\ell = L_{u,\ell} + U_\ell H_\ell^{\mathrm{N}} L_{c,\ell}. \end{aligned} \tag{12.4}$$

These explicit constants give

$$\begin{aligned} |v_\ell(\gamma) - v_\ell(\gamma')| &\le L_{u,\ell}\|\beta - \beta'\|_\infty + U_\ell H_\ell^{\mathrm{N}} L_{c,\ell} \|\alpha - \alpha'\|_\infty \\ &\le L_\ell \|\gamma - \gamma'\|_\infty. \end{aligned} \tag{12.5}$$

Repeated rows and joint parameter restrictions do not invalidate these inequalities; sharper rowwise constants are optional.

13 Polynomial confidence enclosures for policy values

Suppose the statistical construction returns a rational $\mathcal{C}_n \subseteq \Gamma$ and a common measurable coverage event E_n on which $\Gamma_I(p) \subseteq \mathcal{C}_n$. On that event the polytope is nonempty. When nonempty, choose a *fresh* rational $\gamma_{\mathrm{center}} \in \mathcal{C}_n$ by LP. It is distinct from the minimax-fit $\widehat{\gamma}$: the fit uses one common residual tolerance, while (6.2) uses different block tolerances, so $\widehat{\gamma}$ need not belong to $\mathcal{C}_n$.

Compute a certified bound

$$D_n \ge \sup_{\gamma \in \mathcal{C}_n} \|\gamma - \gamma_{\mathrm{center}}\|_\infty.$$

The $2d$ coordinate LPs, for ambient dimension d, give such a radius exactly by taking the largest distance from the center to any coordinate endpoint. Any certified diameter upper bound also suffices because the center belongs to $\mathcal{C}_n$. An empty set is reported as empty and cannot arise on the coverage event under a correctly specified nonempty population model.

For each zero-sum policy, compute at γ_{center} feasible LP bounds $[\underline{v}_\ell, \overline{v}_\ell]$ with gap at most ε_ℓ. Equations (11.7), (11.8), and (12.5) imply that

$$I_{\ell,n} = [\,\underline{v}_\ell - L_\ell D_n,\ \overline{v}_\ell + L_\ell D_n\,] \tag{13.1}$$

contains every value in $\mathcal{W}_I(\ell; p)$ for every declared policy on the same event E_n. Its length is at most

$$\varepsilon_\ell + 2L_\ell D_n. \tag{13.2}$$

This separates LP numerical error from sampling uncertainty. The interval is a certified enclosure, not a claim of sharpness over $\mathcal{C}_n$.

For two policy values, bracket $v_\ell(\gamma_{\text{center}}) - v_{\ell'}(\gamma_{\text{center}})$, then enlarge each side by $(L_\ell + L_{\ell'})D_n$. The numerical gap is at most $\varepsilon_\ell + \varepsilon_{\ell'}$. The same parameter is kept. Selection is immaterial only because each target is the zero-sum utility value, not because policy equilibria are unique.

Under the full-rank statistical condition, on $\Delta \le e_n$, equation (6.6) gives

$$D_n \le \operatorname{diam}_2(\mathcal{C}_n) \le 4\sqrt{R}\, C e_n/\kappa$$

if the exact coordinate radius is used. Indeed each coordinate distance is bounded by the true diameter. Thus (13.2) is an honest root-n enclosure at fixed input, confidence, and positive conditioning. Choose ε_ℓ of the same or smaller order. Weak conditioning enlarges the constants explicitly. All computations for these intervals are polynomial in the explicit tree and rational confidence-polytope input; this makes no succinct- horizon claim.

14 A limit to general outcome contraction at every confidence level

Parameter rates alone do not imply continuity of a policy's entire equilibrium-outcome range. The following example lies inside the declared affine class. Let $\theta \in [-1/4, 1/4]$ and let baseline Nature draw $Y \sim \text{Bernoulli}(1/2 + \theta)$. Baseline utilities are zero. If an action stage is desired, one active player chooses $a \in \{0, 1\}$, still with zero utility, and the same independent fair mixed behavior is selected at every parameter and across markets. The researcher observes (Y, a) completely. The ancillary action supplies no information about θ. The identity measurement is full rank, all complete-history probabilities are at least $1/8$, and Nature probabilities are uniformly positive. The identified parameter is

$$\theta = \mathbb{P}(Y = 1) - 1/2, \qquad \widehat{\theta} = \overline{Y} - 1/2, \qquad \mathbb{E}(\widehat{\theta} - \theta)^2 \le 1/(4n).$$

Ordinary Hoeffding intervals give uniform root-n coverage.

The policy changes only the active player's utility to θa, and the outcome statistic is $W = a$. Its complete range over sequential equilibria is

$$\mathcal{R}(\theta) = \begin{cases} \{0\}, & \theta < 0, \\ [0, 1], & \theta = 0, \\ \{1\}, & \theta > 0. \end{cases} \tag{14.1}$$

Strict preference forces the nonzero-sign cases at every information set. At zero, every behavior is rational and admits fixed-game completely mixed approximations with the corresponding Bayes beliefs; every expected action between zero and one is attained.

Use the joint rational polytope $\alpha = \beta = \theta \in [-1/4, 1/4]$. Its equality transfers the identified chance coefficient to the payoff coefficient, so the example meets the common model even though an ambient-rank test need not be necessary. A dummy second player with utility $-\theta a$ embeds it into a zero-sum game. Its utility value $\max\{0, \theta\}$ is Lipschitz, while the different outcome a has the discontinuous range (14.1). Zero-sum structure does not make every outcome equal to the value.

Theorem 19 (No uniformly contracting honest general-outcome sets)**.** *Fix any $0 < \eta < 1$. Let $\mathcal{D}_n$ be measurable random closed sets with whole-range finite-sample honesty*

$$\inf_{\theta \in [-1/4, 1/4]} \mathbb{P}_\theta\{\mathcal{R}(\theta) \subseteq \mathcal{D}_n\} \ge 1 - \eta. \tag{14.2}$$

For $\theta_n = 1/n$, $n \geq 4$,

$$\mathbb{P}_{\theta_n}\{\mathrm{diam}(\mathcal{D}_n) \geq 1,\ \mathrm{excess}(\mathcal{D}_n, \mathcal{R}(\theta_n)) \geq 1\} \ \geq\ 1 - \eta - \tfrac{1}{2}\sqrt{(1+4/n^2)^n - 1}. \tag{14.3}$$

Here $\mathrm{excess}(A, B) = \sup_{x\in A} \inf_{y\in B} |x - y|$. *Thus Hausdorff error cannot vanish uniformly in probability. The same conclusion holds under asymptotic uniform honesty, with an additional vanishing slack.*

Proof. Let P_0, P_θ be the one-market Bernoulli laws. Their likelihood ratio under P_0 takes values $1 + 2\theta$ and $1 - 2\theta$. Hence

$$\mathbb{E}_0\left[\left(\frac{dP_\theta}{dP_0}\right)^2\right] = 1 + 4\theta^2, \qquad \chi^2(P_\theta^n, P_0^n) = (1+4\theta^2)^n - 1.$$

The product identity follows by independence. Cauchy–Schwarz applied to the centered likelihood ratio gives

$$\mathrm{TV}(P_\theta^n, P_0^n) \leq \tfrac{1}{2}\sqrt{(1+4\theta^2)^n - 1}.$$

An independent ancillary action leaves this distance unchanged. At $\theta_n = 1/n$ the bound tends to zero.

At zero, honesty says the single event $A_n = \{[0,1] \subseteq \mathcal{D}_n\}$ has probability at least $1 - \eta$. The probabilities of that same event under P_0^n and $P_{\theta_n}^n$ differ by at most their total variation, giving the right-hand side of (14.3). On A_n, the reported set contains zero and one, while $\mathcal{R}(\theta_n) = \{1\}$. Its diameter and its excess over the true range are both at least one, proving the result. The excess also lower-bounds the Hausdorff error. Asymptotic uniform honesty repeats the argument with its vanishing coverage slack. There is no intersection of two confidence events, so the proof works for every $0 < \eta < 1$. □

The obstruction is equilibrium indifference, not a discontinuous primitive statistic or vanishing Nature probability. The terminal statistic a is bounded and polynomial. General policy outcomes therefore retain the exact range and confidence-coverage methods; they do not inherit the value contraction theorem.

15 Scope, attribution, and a rational illustration

The results retain every baseline and policy sequential assessment in the declared fixed-tree, positive-Nature class. The generic exact branch includes rank-deficient observation channels and strategically zero baseline histories. The informative branch provides a proved observation restriction, compatibility test, and sharp affine slice for arbitrary finite known private-history trees.

The statistical construction separates three questions. The minimax parameter fit estimates admitted primitives. The outer polytope covers the full sharp parameter fiber. The reconstructed profile has exact feasibility and Bayes consistency, with a computed certificate for all continuation deviations. The latter is not a replacement for exact policy-equilibrium coverage. General counterfactual sets retain exact endpoints and numerical error certificates, while the fast contraction result concerns zero-sum utility values. Comparative effects keep a shared structural parameter and either all policy pairs or an explicitly supplied joint selection relation.

All polynomial-time claims specify an explicit rational input. The complete history tree can be exponentially larger than a compact state/horizon description. Full-history measurement calibration is substantive. Unknown labels, hidden equilibrium mixtures, missing payoff normalization, weak support, or weak rank cannot be removed by reporting a successful optimizer.

The bounds display where those conditions enter. Varying primitive support, arbitrary unencoded utilities, and infinite horizons require different theorems.

As one rational instance of Proposition 9, take

$$\mathsf{Q} = \begin{pmatrix} 7/16 & 1/16 \\ 1/16 & 7/16 \end{pmatrix}, \qquad \theta = (3/4, 5/4), \qquad \xi = 1/2.$$

Then

$$p^a = (4/7, 4/9), \qquad q^a = (16/27, 80/189), \qquad v = 31/63.$$

There are sixteen complete terminal histories, and their minimum mass is $11/972 \geq 1/144 = q_{\min}/9$ with $q_{\min} = 1/16$. The inverse channel and conditional equations recover the same labeled type law and payoff coefficients. These exact arithmetic identities illustrate the general proof; they do not constitute a full run of the QE, LP, or statistical pipeline, and no such execution is claimed here.

The exact SE/QE baseline, compactness, endpoint method, and elementary whole-set coverage are retained from the prior attempt [4]. The general informative fiber and its conditioning, diameter, and actual-assessment certificates supply the additional obligations left there. Sequence form, real quantifier elimination, Brouwer's fixed-point theorem, and rational LP strong duality and polynomial-bit algorithms are classical inputs. The manuscript supplies the reductions, equivalences, and bounds needed for their present use.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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